H-B-491K
September 29, 1993
SUPERSEDING
H-B-491J
Mar. 21, 1986

FEDERAL SPECIFICATION

BRUSH, PAINT, SASH AND TRIM

The General Services Administration has authorized the use of this federal specification, for all federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 SCOPE. This specification covers oval and angular sash tool paint brushes.

1.2 Classification. The brushes shall be of the following types, classes and sizes as specified (see 6.2). (Size denotes width or diameter and length of bristle outside of ferrule.):

Type I - Oval

Class 1 - Bristle
Class 2 - Synthetic Filament

Size,	mm (inches)
6	22 by 18 (7/8) by (11/16)
8	25 by 21 (1) by (13/16)
10	29 by 24 (1-1/8) by (15/16)
12	32 by 27 (1-1/4) by (1-1/16)

Type II - Angular

Class 1 - Bristle
Class 2 - Synthetic

Size,	mm (inches)
5	25 by 56 (1) by (2-3/16)
7	38 by 56 (1-1/2) by (2-3/16)
9	51 by 62 (2) by (2-7/16)
11	64 by 68 (2-1/2) by (2-11/16)
13	76 by 75 (3) by (2-15/16)

2. APPLICABLE DOCUMENTS.

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extent specified herein.

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951 -Synthetic Filaments for Brushes, General Specification for
PPP-B-1880 -Brushes; preparation for Delivery of
TT-T-291 -Thinner, Paint, Mineral Spirits, Regular and odorless

Military Standards:

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes

Commercial Item Descriptions:

A-A-857 - Thinner, Dope and Lacquer

Federal Regulations:

29 CFR 110.1200 -Hazard Communication Standard

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from GSA, Specification Section (3FBP-W), Suite 8100, 470 L'Enfant Plaza, S.W., Washington, DC 20407)

National Motor Freight Traffic Association, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the ATA, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Chemical Society (ACS) Specifications:

Reagent Chemicals (Latest Edition)
Hydrochloric acid

(Application for copies should be addressed to the American Chemical Society, 1122 Sixteenth Street, N.W., Washington, DC 20036.)

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American Society for Testing and Materials:

ASTM D13 -Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS.

3.1 Materials and components. Materials and components shall conform to the requirements specified herein.

3.1.1 Bristle (Class 1). The bristle for Class 1 brushes shall be untreated natural hog bristle. The bristle shall be boiled or steamed to prevent twisting or curling. Recycled bristle will not be accepted. The bristle shall pass the color test specified in 4.4.1 and the straightness test specified in 4.4.5. A certificate of compliance from the supplier may be accepted that the bristle meets the stiffness, straightness, and sterilization requirements.

3.1.2 Synthetic filaments (Class 2). The synthetic filament for Class 2 brushes shall be an organic polymeric material conforming to H-S-951, class N, type I or Class E, type I tapered filaments meeting the size designation of Table II. All filaments shall be tipped to a depth of not less than 5mm. (3/16 inch). After tipping, 20 percent, by weight, of the top-sized filaments, shall be flagged. (See 6.3.1)

3.1.3 Ferrule. The ferrule shall be made of corrosion resistant metal, or metal made corrosion resistant by a coating which shall not wrinkle, blister, separate, or become soft or dull when tested in accordance with 4.4.3.

3.1.4 Handle. The handle shall be smoothly finished close grained hardwood or a synthetic material with all sprues and runners removed. The synthetic handle shall not crack or break when tested as in 4.4.4, and shall show no perceptible change when tested as in 4.4.3.

3.1.5 Filler strips. Filler strips shall be made of wood, metal, or plastic and shall not soften or otherwise change when tested as in 4.4.3. Cardboard filler strips that do not soften or otherwise change when tested as in 4.4.3 may be used in machine made brushes.

3.1.6 Setting Compound. The setting compound shall be an epoxy cement which shall withstand the test specified in 4.4.3 without softening or deteriorating, and without loosening of bristle or filament in the compound.

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3.2 Construction and physical requirements.

3.2.1 Ferrule. The type I ferrule metal shall have a minimum wall thickness of 0.4 mm. (0.015 inch). The type II ferrule metal shall have a minimum wall thickness of 0.25 mm (0.01 inch) and shall be formed with an interlocked, lock-seamed, soldered, or welded endlap, and shall have a groove at the base to aid in controlling the flow of the setting compound.

3.2.2 Set. The butt ends of the bristles or filaments shall be set in a setting compound conforming to 3.1.6. When tested, as in 4.4.3.3, there shall be no more than a total of five (5) bristles or filaments pulled loose during the tensile pull.

3.2.3 Assembly. The ferrule shall be fastened to the handle, as follows, using corrosion resistant nails with a minimum length of 9 mm. (3/8 inch) and a shaft diameter of 1 - 2 mm. (0.04-0.06 inch):

(a) For oval (Type I) brushes, there shall be a minimum of three nails evenly spaced around the ferrule.

(b) For Type II, size 5 brushes, there shall be a minimum of three nails used, with two nails evenly spaced on one side and one nail approximately centered on the opposite side.

(c) For Type II, sizes 7, 9, 11, and 13 brushes, there shall be a minimum of four nails used, with two nails, evenly spaced, on each side of the brush.

3.2.4 Trim. Brushes shall be cupped or machine chiseled with a minimum chisel of 6 mm. (1/4 inch). If machine chiseled, the ends shall be sanded. The flagged ends of the top size bristles shall be preserved. The ends of synthetic filament shall be sanded to a fine soft edge.

3.3 Composition and physical requirements.

3.3.1 Class I. The bristle shall be 60 percent top size and blended in accordance with Table I. (Dressed bristle, described as 60 percent top size, denotes that bundle consists of 60 percent bristle measuring 2 mm. (1/16 inch) or more above the length specified to 3 mm. (1/8 inch) below the length specified; 10 percent will be 9.5 mm (3/8 inch) below the length specified; 10 percent 9.5 mm (3/8 inch) to not more than 12.7 mm (1/2 inch) below specified length but including a few unaviodable shorts and blended in accordance with Table I.

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TABLE I. Physical requirements for type I, class 1 and 2 brushes

Characteristic:	Brush size 6	8	10	12
Width of brush inside ferrule, mm., (in.), min.	21 (27/32)	25 (31/32)	28 (1-3/32)	31 (1-7/32)
Brush thickness inside ferrule, mm., (in.), min.	17 (21/32)	20 (25/32)	23 (29/32)	26 (1-1/32)
Bristle/filament length outside ferrule, mm., (in.), Min.	56 (2-3/16)	62 (2-7/16)	68 (2-11/16)	75 (2-15/16)
Diameter of handle at ferrule, mm., (in.), Min.	19 (3/4)	23 (29/32)	26 (1-1/32)	27 (1-1/16)
Weight of bristles per class 1 brush, grams, Min.	10	15	20	29
Weight of filament per class 2 brush, grams, Min.	11	17	23	33
Blend of bristles, percentage and length, mm., (inches)	100% 64 (2-1/2)	20% 64 (2-1/2)	5% 64 (2-1/2)	5% 70 (2-3/4)
	------	80% 70 (2-3/4)	15% 70 (2-3/4)	15% 76 (3)
	------	------	80% 76 (3)	80% 83 (3-1/4)
Blend of filament, percentage and length, mm., (in.):				
Filament Size: 9-5	100% 64 (2-1/2)	70% 70 (6.99)	60% 76 (7.62)	50% 83 (8.26)
12-8	------	30% 64 (2-1/2)	20% 64 (2-1/2)	20% 70 (2-3/4)
12-8	------	------	20% 70 (2-3/4)	20% 76 (3)
12-8	------	------	------	10% 83 (3-1/4)

Bristle 76 mm (3-inches) and longer shall be 20% stiff and 80% soft. Bristle shorter than 76 mm (3 inches) shall be 100% soft.

3.3.2 **Type** II. Synthetic filament conforming to 3.1.2 shall be blended as specified in table II, with a tolerance of plus or minus 3 percent for each length and stiffness specified.

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TABLE II. Physical requirements for type II, class 1 and 2 brushes.

Characteristic:	Brush Size 5	7	9	11	13
Width of brush inside ferrule, mm, (in), Min.	25 (1)	38 (1-1/2)	51 (2)	64 (2-1/2)	76 (3)
Brush thickness inside ferrule, mm., (in), Min.	10 (3/8)	11 (7/16)	13 (1/2)	16 (5/8)	18 (11/16)
Bristle/filament length outside ferrule, mm., (in), Min.	56 (2-3/16)	56 (2-3/16)	62 (2-7/16)	68 (2-11/16)	24 (2-15/16)
Class I: Weight of bristles per class 1 brush, grams, Min.	9	15	21	27	40
Class II: Weight of filament per class 2 brush, grams, Min.	9	15	23	33	50
Class I: Blend of bristles, percentage and length, mm., (in.), Min.	100% 64 (2-1/2)	100% 64 (2-1/2)	20% 64 (2-1/2)	5% 64 (2-1/2)	5% 70 (2-3/4)
	------	------	80% 70 (2-3/4)	15% 70 (2-3/4)	15% 76 (3)
	------	------	------	80% 76 (3)	80% 83 (3-1/4)
Class II: Blend of filament, percentage and length, mm., (in):					
Filament Size: 9-5	30% 57 (2-1/4)	30% 57 (2-1/4)	------	------	------

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TABLE II. Physical requirements for type II, class 1 and 2 brushes.

Characteristic:		Brush Size 5	7	9	11	13
Filament Size:	9-5	70% 64 (2-1/2)	70% 64 (2-1/2)	70% 70 (2-3/4)	60% 76 (3)	50% 83 (3-1/4)
	12-8	------	------	30% 64 (2-1/2)	20% 64 (2-1/2)	10% 83 (3-1/4)
	12-8	------	------	------	20% 70 (2-3/4)	20% 76 (3)
	12-8	------	------	------	------	20% 70 (2-3/4)
Ferrule Height:						
Long Side, mm., (in), +/- 1/8"		38 (1-1/2)	45 (1-3/4)	45 (1-3/4)	48 (1-7/8)	48 (1-7/8)
Short Side, mm., (in), +/- 1/8"		33 (1-5/16)	37 (1-7/16)	37 (1-7/16)	33 (1-5/16)	30 (1-3/16)

Bristle 83 mm. (3-1/4 inches) and longer shall be 20% stiff and 80% soft. Bristle shorter than 83 mm. (3-1/4 inches) shall be 100 % soft.

3.4 **Marking for identification.** Unless otherwise specified (6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number (NSN).

3.5 **Workmanship.** The finished brushes shall be clean, well made, and free from any defect which may affect serviceability or appearance. There shallbe no sharp edges, splinters, burrs, or other imperfections. The brushing part shall be trimmed evenly and uniformly with the flagged ends preserved to the finished length outside the ferrule.

4. QUALITY ASSURANCE PROVISIONS.

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

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4.2 Inspection of preparation for delivery. An examination shall be made to determine compliance with 3.3 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 Inspection of the end item. Lots shall be assembled and randomly sampled in accordance with MIL-STD-105, inspection level II, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles or filaments
Ferrules
Handles
Construction
Set
Trim
Workmanship
Identification markings

Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 Sampling for testing of the end item. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4. Testing procedures. Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 Color test. Samples of bristle shall be soaked in boiling HCl (ACS Reagent Grade, 5 percent by volume in distilled water) for 5 minutes. The solution shall show no color other than green. Any other color shall be indicative of coloring matter.

4.4.2 Weight of bristle or filament. After removal of the ferrule, the setting compound shall be removed from the bristle or filament by any practical means which does not injure the bristle or filament. The weight of bristles or filaments shall be determined for compliance with physical requirements.

4.4.3 Solvent Resistance:

4.4.3.1 Handle (Organic coating). Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a mineral spirits conforming to Federal Specification TT-T-291.

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After 30 minutes, the handle shall be removed from the mineral spirits and inspected for wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and inspected for wrinkling, blistering, or separation of the coating.

C. The brush handle shall be dried for 2 hours and the organic coating shall be examined for softening or dulling.

4.4.3.2 **Base material, ferrule, filler strips, synthetic handles.** The brush shall be tested for solvent resistance using a thinner conforming to Commercial Item Description A-A-857.

(**NOTE:** The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, plating/coating on the ferrule, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in the solvent for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. **Handle and ferrule.** As soon as the brush is removed from the solvent, the organic coating on the ferrule shall be examined for wrinkling, blistering, or separation. The handle shall be examined for any deleterious effect by the solvent. The brush shall then be dried for 2 hours and examined for softening or dulling of the ferrule coating or deterioration of the handle base material.

B. **Filler strips.** When the brush is removed from the solvent, filler strips shall be examined for softening, swelling, or other deleterious change.

C. **Setting compound.** After removal of the brush from the solvent, the setting compound shall be examined for softening, loosening of bristle, and other solvent effects. After drying one hour, the brush shall be suspended by the handle, in a holding device, and a group of bristles, 6 mm (1/4 inch) in diameter, shall be given a tensile pull of 4.5 kg. (10 lbs) away from the ferrule, (type I brushes only). Repeat this tensile pull test, at random places, one time for each 25 mm (1 inch) of brush width. After thoroughly drying the brush, remove the

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specification covers a general purpose, ... scope. This

ferrule and examine the setting compound for deterioration, bristle loosening, or other solvent damage.

4.4.3.3 Filler strips. When the brush is removed from the solvent, filler strips shall be examined for softening, swelling, or other deleterious change.

After thoroughly drying the brush, remove the ferrule and examine the setting compound for deterioration, bristle/filament loosening, or other solvent damage.

4.4.4 Drop test. Remove bristle (or synthetic filament) and ferrule. Synthetic handles shall be dropped twice from a height of 3-meters (10 feet) onto a concrete surface and examined for compliance with 3.1.4 and 3.2.3.

4.4.5 Straightness. Place assembled brush head so that the bristle is immersed in 60 degree C. water for five (5) minutes. Shake out excess water. Hang brush with bristle/filament down to dry completely. Measure flare by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Disregard stray bristles/filaments.

4.4.6 Tipping and flagging of filament. The depth of tipping and flagging shall be measured with a ruler and a 7-power magnifying glass.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (See 6.2).

6. NOTES

6.1 Intended use. Brushes made with natural hog bristle (Class I) are recommended for application of solvent borne paints and coatings. Brushes made with synthetic filaments (Class II) are recommended for application of water borne paints and coatings.

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6.2 **Ordering data.** Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).
(d) Marking required (see 5.1).

6.3 **Definitions.**

6.3.1 **Flagging.** Flagging is a mechanical process to split filament ends to simulate the natural flag of hog bristle.

6.3.2 **Tipping.** Tipping is a process of grinding or sanding filament tips to produce a fine, soft, even tip.

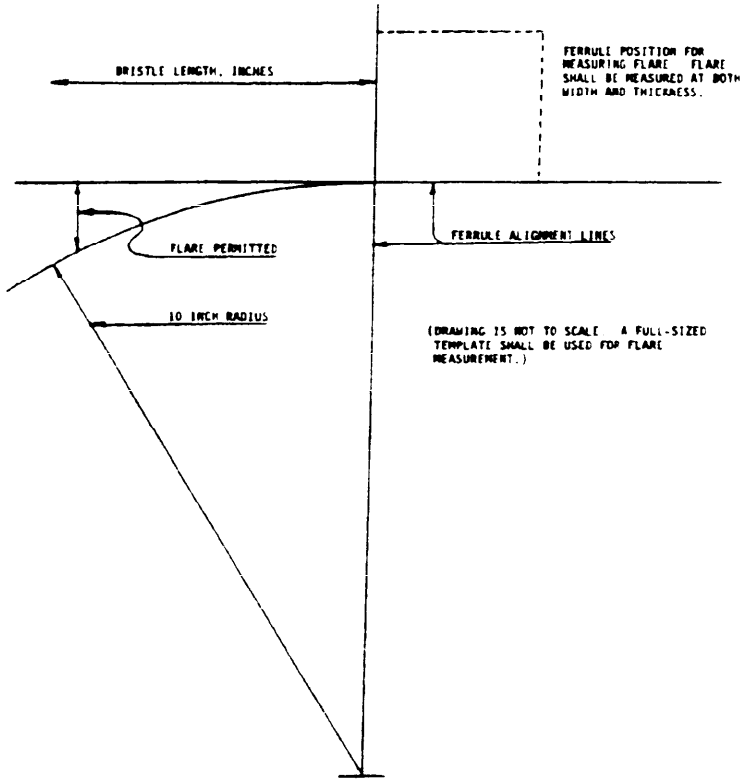


Figure 1. Flare Measurement

PREPARING ACTIVITY
ACTIVITY

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